

Joana Konadu Owusu

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Location: Laramie, Wyoming

Objective/ Summary

I am a PhD Computer Science student supervised by Dr. Shivanand Venkanna Sheshappanavar. My research entails VLM design and improvements for Transportation, Healthcare and Agriculture applications. I am proficient in Computer Vision and Deep Learning techniques, and I am seeking to launch my career as a Research Scientist where I can apply my expertise to develop impactful solutions.

Education

Doctor of Philosophy (PhD) – Computer Science University of Wyoming	June 2026 – May 2028 (Exp. Grad Date) GPA: 4.0
Master of Science (MSc) – Computer Science University of Wyoming	Aug 2024 – May 2026 GPA: 4.0
Bachelor of Science (BSc) – Computer Science & Engineering University of Mines & Technology	Sep 2016 – Aug 2020 CGPA: 3.61

Awards & Honors

CVPR 2026 Broadening Participation Grant	Jun 2026
• Competitive travel award covering conference registration and lodging for the full CVPR 2026 conference.	

Research Projects

IDTCount: Counting Every Instance Once in Open Vocabulary Video Counting (Lead)	May 2026 – Aug 2026
• Built a learned re-identification module and association mechanism inside SAM3's video tracking loop. • Designed a ReID projection head trained via supervised contrastive learning for compact identity representation • Submitted to WACV 2027 (currently under review)	
SAM3Count for Zero-Shot Open Vocabulary Counting in Images and Videos (Lead)	Nov 2025 – Mar 2026
• Built zero-shot counting model targeting counting failures caused by dense scenes in images and videos • Designed an ROI and adaptive tiling preprocessor and an identity layer over SAM3 to maintain consistent identities • Achieved SOTA results across image and video benchmarks tested (over 13% decrease in MAE set). • Accepted for Oral presentation at CVPR Workshop 2026.	
MedQA LLMs (Lead)	Oct 2025 – Nov 2025
• Fine-tuned GPT-2, LLaMA-3.2 1B, and Gemma with LORA on MedQuAD to perform comparative analysis and evaluations with the aim to deduce which LLM is best for Med QA • Evaluated new models with F1, BERTScore, LLM-as-judge (Ollama) and human evaluations with LLaMA being the highest scorer of F1: 33.38%, BERTScore: 62.55%, human: 4.2/5	
Smart Glasses for the Visually Impaired (Lead)	Jan 2025 – Sep 2025
• Engineered an AI-powered smart glasses to assist visually impaired in scene analysis and object identification • Built a cross-platform mobile application to serve as the primary user interface and communication hub to the user • Developed a computer vision and LLM model to process sensory data and provide descriptive feedback to the user	
GPT2 from Scratch (Lead)	Sep 2025 – Nov 2026
• Built an end-to-end GPT model with tokenization, attention, chain of thought reasoning and RAG techniques. • Fine-tuned to follow instructions on instruction–response pairs with logged generation behavior across temperatures.	

Work Experience

Graduate Research Assistant	Aug 2024 – Present
University of Wyoming WY, USA	

Supervisor: Dr.Shivanand Venkanna Sheshappanavar

- Improve and develop VLMs (Image, Video and 3D) for better identification and tracking in assistive technologies, healthcare and autonomous driving.
- Utilize HPC to incorporate techniques that optimize and enhance the efficiency and accuracy of these models.
- Assist the REU NSF funded summer program for mentoring undergraduate students in their research journey.

Information Technology Coordinator

Nov 2021 – July 2024

Ghana Education Service |Kumasi, Ghana

- Administered and maintained the school's database systems and networks while maximizing the information technology tools and systems' efficiency and effectiveness
- Performed Network and System administration tasks on Windows Server and SAP software.
- Reviewed schools' system and network performance to make proactive adjustments based on established IT policies.

Publications

Joana Konadu Owusu, Shivanand Venkanna Sheshappanavar, "SAM3Count for Zero-Shot Open Vocabulary Counting in Images and Videos", (Oral), Computer Vision and Pattern Recognition Workshop Proceedings (CVPR) 2026. [\[PDF\]](#)

Joana Konadu Owusu, Shivanand Venkanna Sheshappanavar, "IDTCount: Counting Every Instance Once in Open Vocabulary Video Counting", Under Review, Winter Conference on Applications of Computer Vision (WACV) 2027. [\[PDF\]](#)

Joana Konadu Owusu, Shivanand Venkanna Sheshappanavar, "Object Counting Across Modalities: Taxonomies, Benchmarks, Applications, and Open Challenges", Under Review, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2027. [\[PDF\]](#)

Presentations

"Open-Vocabulary Object Counting: From Zero-Shot Detection to Identity-Aware Video Counting" (Talk), Mountain West AI & HPC Summit, University of Wyoming, Aug 2026.

"SAM3Count for Zero-Shot Open Vocabulary Counting in Images and Videos" (Poster), Women in Computer Vision Workshop @ CVPR 2026, Colorado Convention Center, Jun 2026.

"Vehicle Tracking in Adverse Weather" (Poster), 2025 NSF REU Site: Design, Create, and Innovate 3-Dimensional User Interfaces to Improve Human Sensory and Motor Performance in Virtual Environments (HUMAN MOVE), University of Wyoming, EERB Atrium, Jul 2025.

"Enhancing Road Safety in Blowing Snow Conditions: Advanced Detection of Lanes, Reflectors, and Vehicles" (Poster), NSF IUCRC AMRITA Planning Meeting, University of Wyoming, May 2025.

Skills & Certifications

Technical Languages: Python, SQL, Java, JavaScript.

Operating Systems: Windows, Linux, Mac OS.

Certifications: LinkedIn Full Stack Software Developer, Microsoft Azure Fundamentals, IBM Enterprise Design Thinking Practitioner

Tools: GitHub, HPC, SLURM, Jupyter Notebook, Pytorch, CUDA, RAG, Node JS, React, MySQL, MongoDB, Microsoft Azure, AWS (SageMaker, Snow Flake, S3, IAM, CloudWatch, EC2, Lambda), ML Ops, CI/CD, REST API, Raspberry PI

Soft Skills: Mission-Driven, Collaborative, Strong Communicator, Eager to Learn, Passion for Growth.

Extracurricular & Leadership Activities

Volunteer Software Development Specialist

Inclusive Tech Group

Nov 2020 – to date

- Member of the core team that plans, host and organize events, training sessions, conferences, fairs and hackathons to advocate for Disability Inclusion in the technology space.
- Collaborate with members of the product development team to develop assistive technologies for persons with disabilities including mobile apps, web platforms and hardware IoT devices.